A APPENDIX

A.1 Proof of Lemma 1

In order to demonstrate the convergence of our update scheme, we make the following assumptions:

- Lipschitzian gradient: $f(\boldsymbol{x})$ is L-Lipschitz smooth, i.e., $||\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})|| \leq L||\boldsymbol{x} - \boldsymbol{y}||$, $\forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^d$.
- Bounded variance: the variance of the stochastic gradient, denoted by $\boldsymbol{\xi}$, is bounded such that $\mathbb{E}[\|\boldsymbol{\xi}\|^2] \leq \sigma^2$.

Due to the L-Lipschitz condition satisfied by $f(\boldsymbol{x})$, the derivative of the formula $||\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})|| \leq L||\boldsymbol{x} - \boldsymbol{y}||$ with respect to $\boldsymbol{x}$ yields $\nabla^2 f(\boldsymbol{x}) \leq L$. Therefore, we can obtain the second-order Taylor expansion of $f(\boldsymbol{x})$ around $\nabla^2 f(\boldsymbol{x}) \leq L$.

$$f(\boldsymbol{y}) \leq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^T(\boldsymbol{y} - \boldsymbol{x}) + \frac{L}{2}||\boldsymbol{y} - \boldsymbol{x}||^2 \tag{1}$$

Therefore, we can deduce the following proof.

$$\begin{aligned}
&\mathbb{E}f(\boldsymbol{x}_{t+1}) - \mathbb{E}f(\boldsymbol{x}_t) \\
&\leq \mathbb{E}\left\langle \boldsymbol{x}_{t+1} - \boldsymbol{x}_t, \nabla f(\boldsymbol{x}_t)\right\rangle + \frac{L}{2}\mathbb{E}||\boldsymbol{x}_{t+1} - \boldsymbol{x}_t||^2 \\
&= \mathbb{E}\left\langle -\gamma(\nabla f(\boldsymbol{x}_t) - p\xi_t), \nabla f(x_t)\right\rangle + \frac{L\gamma^2}{2}\mathbb{E}||\nabla f(\boldsymbol{x}_t) - p\boldsymbol{\xi}_t||^2 \quad (\nabla g(\boldsymbol{x}) = \nabla f(\boldsymbol{x}) - p \cdot \boldsymbol{\xi} \text{ and } \boldsymbol{x_{t+1}} = \boldsymbol{x_t} - \gamma\nabla g(\boldsymbol{x_t})) \\
&\leq -\gamma\mathbb{E}\left\langle \nabla f(\boldsymbol{x}_t) - p\xi_t, \nabla f(x_t)\right\rangle + \frac{L\gamma^2}{2}\mathbb{E}||\nabla f(\boldsymbol{x}_t) - p\boldsymbol{\xi}_t||^2 \\
&\leq -\gamma\mathbb{E}||\nabla f(\boldsymbol{x}_t)||^2 + \frac{L\gamma^2}{2}\mathbb{E}||\nabla f(\boldsymbol{x}_t)||^2 + \frac{L\gamma^2}{2}p^2\mathbb{E}\boldsymbol{\xi}_t^2 \quad (\mathbb{E}[\boldsymbol{\xi}] = 0) \\
&\leq (-\gamma + \frac{L\gamma^2}{2})\mathbb{E}||\nabla f(\boldsymbol{x}_t)||^2 + \frac{L\gamma^2}{2}p^2\sigma^2 \quad (\mathbb{E}[||\boldsymbol{\xi}||^2] \leq \sigma^2)
\end{aligned}$$

Using the method of telescoping sum, we can obtain the following equations.

$$\begin{aligned}
&\mathbb{E}f(\boldsymbol{x}_T) - \mathbb{E}f(\boldsymbol{x}_0) \leq (\frac{L\gamma^2}{2} - \gamma)\sum_{t=0}^{T-1}\mathbb{E}||\nabla f(\boldsymbol{x}_t)||^2 + \frac{L\gamma^2}{2}p^2\sigma^2 T \\
&(\gamma - \frac{L\gamma^2}{2})\sum_{t=0}^{T-1}\mathbb{E}||\nabla f(\boldsymbol{x}_t)||^2 \leq \mathbb{E}f(\boldsymbol{x}_0) - \mathbb{E}f(\boldsymbol{x}_t) + \frac{L\gamma^2}{2}p^2\sigma^2 T \\
&(\gamma - \frac{L\gamma^2}{2})\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}||\nabla f(\boldsymbol{x}_t)||^2 \leq \frac{\mathbb{E}f(\boldsymbol{x}_0) - \mathbb{E}f(\boldsymbol{x}_t)}{T} + \frac{L\gamma^2}{2}p^2\sigma^2 \\
&\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}||\nabla f(\boldsymbol{x}_t)||^2 \leq \frac{2L(\mathbb{E}f(\boldsymbol{x}_0) - \mathbb{E}f(\boldsymbol{x}_t))}{T} + p^2\sigma^2
\end{aligned}$$

Lemma 1 suggests that by setting the probability of selecting hybrid batches for training to $p = 1/\sqrt{T}$, we achieve a convergence rate of $O(1/T)$, equivalent to that of full-vertex batch training.

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}||\nabla f(\boldsymbol{x}_t)||^2 \leq \frac{2L(\mathbb{E}f(\boldsymbol{x}_0) - \mathbb{E}f(\boldsymbol{x}_t)) + \sigma^2}{T} \tag{2}$$

Setting $p = 1/\sqrt[4]{T}$, however, leads to a convergence rate of $O(1/\sqrt{T})$, akin to mini-batch training.

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}||\nabla f(\boldsymbol{x}_t)||^2 \leq \frac{2L(\mathbb{E}f(\boldsymbol{x}_0) - \mathbb{E}f(\boldsymbol{x}_t))}{T} + \frac{\sigma^2}{\sqrt{T}} \tag{3}$$

Moreover, we can ensure convergence by setting the learning rate γ.

$$
\begin{aligned}
(\gamma - \frac{L\gamma^2}{2})\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}||\nabla f(\boldsymbol{x_t})||^2 &\leq \frac{\mathbb{E}f(\boldsymbol{x_0}) - \mathbb{E}f(\boldsymbol{x_t})}{T} + \frac{L\gamma^2}{2}p^2\sigma^2 \\
\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}||\nabla f(\boldsymbol{x_t})||^2 &\leq \frac{2(\mathbb{E}f(\boldsymbol{x_0}) - \mathbb{E}f(\boldsymbol{x_t}))}{T(2\gamma - L\gamma^2)} + \frac{L\gamma^2 p^2\sigma^2}{2\gamma - L\gamma^2} \quad (\gamma < \frac{2}{L}) \\
\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}||\nabla f(\boldsymbol{x_t})||^2 &\leq \frac{2(\mathbb{E}f(\boldsymbol{x_0}) - \mathbb{E}f(\boldsymbol{x_t}))}{T\gamma^2(\frac{2}{\gamma} - L)} + \frac{Lp^2\sigma^2}{\frac{2}{\gamma} - L}
\end{aligned}
$$

We make $\frac{L}{\frac{2}{\gamma}-L} = \frac{1}{\sqrt{T}}$, i.e., $\gamma = \frac{2}{L(\sqrt{T}+1)}$, and then we can get:

$$
\begin{aligned}
\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}||\nabla f(\boldsymbol{x_t})||^2 &\leq \frac{2(\mathbb{E}f(\boldsymbol{x_0}) - \mathbb{E}f(\boldsymbol{x_t}))}{T\gamma^2(\frac{2}{\gamma} - L)} + \frac{Lp^2\sigma^2}{\frac{2}{\gamma} - L} \\
&\leq \frac{2(\mathbb{E}f(\boldsymbol{x_0}) - \mathbb{E}f(\boldsymbol{x_t}))}{L\sqrt{T}\gamma^2} + \frac{p^2\sigma^2}{\sqrt{T}} \\
&\leq \frac{(\mathbb{E}f(\boldsymbol{x_0}) - \mathbb{E}f(\boldsymbol{x_t}))L(\sqrt{T}+1)^2}{2\sqrt[3]{T}} + \frac{p^2\sigma^2}{\sqrt{T}} \\
&\leq \frac{\Omega}{2\sqrt[3]{T}} + \frac{\Omega}{2T} + \frac{2p^2\sigma^2 + \Omega}{2\sqrt{T}} \quad (\Omega = (\mathbb{E}f(\boldsymbol{x_0}) - \mathbb{E}f(\boldsymbol{x_t}))L)
\end{aligned}
$$

Setting $\gamma = \frac{2}{L(\sqrt{T}+1)}$, we can get a convergence rate of $O(\frac{1}{\sqrt{T}})$, akin to mini-batch training.

A.2 Proof of Lemma 2

Motivated by K-AVG [1], we provide the convergence proof of our updating schema under Asynchronous Model Controller. Continuing A.1, we give the formula for M workers with our updating schema. Here, $\tilde{x}_{\alpha+1}$ represents the average model of M workers at step $\alpha+1$. From $\tilde{x}_\alpha$ to $\tilde{x}_{\alpha+1}$, each worker takes K iterations to update their local models.

$$
\tilde{x}_{\alpha+1} = \frac{1}{M}\sum_{j=1}^{M} x^j_{\alpha+K} = \frac{1}{M}\sum_{j=1}^{M}\left[x^j_\alpha - \gamma\sum_{t=0}^{K-1}\left(\nabla f(x^j_{\alpha+t}; D^j_t) - p^j_{\alpha+t}\xi^j_{\alpha+t}\right)\right]
$$

By using Equation 1, we can get:

$$
\begin{aligned}
&f(\tilde{x}_{\alpha+1}) - f(\tilde{x}_\alpha) \leq \langle\nabla f(\tilde{x}_\alpha), \tilde{x}_{\alpha+1} - \tilde{x}_\alpha\rangle + \frac{L}{2}\|\tilde{x}_{\alpha+1} - \tilde{x}_\alpha\|_2^2 \\
&\leq -\langle\nabla f(\tilde{x}_\alpha), \gamma\frac{1}{M}\sum_{j=1}^{M}\sum_{t=0}^{K-1}(\nabla f(x^j_{\alpha+t}; D^j_t) - p^j_{\alpha+t}\xi^j_{\alpha+t})\rangle + \frac{L}{2}\|\gamma\frac{1}{M}\sum_{j=1}^{M}\sum_{t=0}^{K-1}(\nabla f(x^j_{\alpha+t}; D^j_t) - p^j_{\alpha+t}\xi^j_{\alpha+t})\|_2^2
\end{aligned}
$$

Then we take the expectation of it:

$$
\mathbb{E}f(\tilde{x}_{\alpha+1}) - \mathbb{E}f(\tilde{x}_\alpha) \leq -\gamma\sum_{t=0}^{K-1}\mathbb{E}\langle\nabla f(\tilde{x}_\alpha), \nabla f(x^j_{\alpha+t})\rangle + \frac{L\gamma^2}{2M^2}\mathbb{E}||\sum_{j=1}^{M}\sum_{t=0}^{K-1}\nabla f(x^j_{\alpha+t}; D^j_t) - p^j_{\alpha+t}\xi^j_{\alpha+t}||_2^2 \tag{4}
$$

First, we derive the former part $-\gamma\sum_{t=0}^{K-1}\mathbb{E}\langle\nabla f(\tilde{x}_\alpha), \nabla f(x^j_{\alpha+t})\rangle$ of Equation 4.

$$
-\gamma\sum_{t=0}^{K-1}\mathbb{E}\langle\nabla f(\tilde{x}_\alpha), \nabla f(x^j_{\alpha+t})\rangle \leq -\frac{(K+1)\gamma}{2}||\nabla f(\tilde{x}_\alpha)||_2^2 - \frac{\gamma}{2}\sum_{t=1}^{K-1}\mathbb{E}||\nabla f(x^j_{\alpha+t})||_2^2 + \frac{L^2\gamma}{2}\sum_{t=1}^{K-1}\mathbb{E}||x^j_{\alpha+t} - \tilde{x}_\alpha||_2^2 \tag{5}
$$

We derive the last part $\frac{L^2\gamma}{2}\sum_{t=1}^{K-1}\mathbb{E}||x^j_{\alpha+t} - \tilde{x}_\alpha||_2^2$ of Equation 5.

$$
\begin{aligned}
\mathbb{E}||x^j_{\alpha+t} - \tilde{x}_\alpha||_2^2 &= \gamma^2\mathbb{E}||\sum_{i=0}^{t-1}\nabla f(x^j_{\alpha+i}; D^j_i) - p^j_{\alpha+i}\xi^j_{\alpha+i}||_2^2 \\
&\le t\gamma^2\mathbb{E}\sum_{i=0}^{t-1}||\nabla f(x_{\alpha+i}; D^j_i) - p^j_{\alpha+i}\xi^j_{\alpha+i}||_2^2 \\
&\le t\gamma^2\mathbb{E}\sum_{i=0}^{t-1}||\nabla f(x_{\alpha+i}; D^j_i)||_2^2 + t\gamma^2\mathbb{E}\sum_{i=0}^{t-1}||p^j_{\alpha+i}\xi^j_{\alpha+i}||_2^2 \\
&\le t\gamma^2\mathbb{E}\sum_{i=0}^{t-1}||\nabla f(x_{\alpha+i}; D^j_i)||_2^2 + t\gamma^2\check{p}^2\sum_{i=0}^{t-1}\mathbb{E}||\xi^j_{\alpha+i}||_2^2 \\
&\le t\gamma^2\mathbb{E}\sum_{i=0}^{t-1}||\nabla f(x_{\alpha+i}; D^j_i)||_2^2 + t^2\gamma^2\check{p}^2\sigma^2 \\
&= t\gamma^2\mathbb{E}\sum_{i=0}^{t-1}||\nabla f(x^j_{\alpha+i}; D^j_i) - \nabla f(x^j_{\alpha+i})||_2^2 + t\gamma^2\mathbb{E}\sum_{i=0}^{t-1}||\nabla f(w^j_{\alpha+i})||_2^2 + t^2\gamma^2\check{p}^2\sigma^2 \\
&\le t^2\gamma^2\theta + t^2\gamma^2\check{p}^2\sigma^2 + t\gamma^2\mathbb{E}\sum_{i=0}^{t-1}||\nabla f(w^j_{\alpha+i})||_2^2 \\
&\le t^2\gamma^2(\theta + \check{p}^2\sigma^2) + t\gamma^2\mathbb{E}\sum_{i=0}^{t-1}||\nabla f(w^j_{\alpha+i})||_2^2
\end{aligned}
$$

We substitute the above equation into Equation 5 and obtain the following results.

$$
\begin{aligned}
-\gamma\sum_{t=0}^{K-1}\mathbb{E}\langle\nabla f(\tilde{x}_\alpha), \nabla f(x^j_{\alpha+t})\rangle \le &-\frac{(K+1)\gamma}{2}(1 - \frac{L^2\gamma^2K(K-1)}{2(K+1)})||\nabla f(\tilde{x}_\alpha)||_2^2 - \frac{\gamma}{2}(1 - \frac{L^2\gamma^2(K+1)(K-2)}{2})\sum_{t=1}^{K-1}\mathbb{E}||\nabla f(x^j_{\alpha+t})||_2^2 \\
&+ \frac{L^2\gamma^3(K-1)K(2K-1)}{12}(\theta + \check{p}^2\sigma^2)
\end{aligned}
$$

Second, we derive the latter part $\frac{L\gamma^2}{2M^2}\mathbb{E}||\sum_{j=1}^{M}\sum_{t=0}^{K-1}\nabla f(x^j_{\alpha+t}; D^j_t) - p^j_{\alpha+t}\xi^j_{\alpha+t}||_2^2$ of Equation 4.

$$
\begin{aligned}
\frac{L\gamma^2}{2M^2}\mathbb{E}||\sum_{j=1}^{M}\sum_{t=0}^{K-1}\nabla f(x^j_{\alpha+t}; D^j_t) - p^j_{\alpha+t}\xi^j_{\alpha+t}||_2^2 &\le \frac{LK\gamma^2}{2M^2}\mathbb{E}\sum_{t=0}^{K-1}||\sum_{j=1}^{M}\nabla f(x^j_{\alpha+t}; D^j_t) - p^j_{\alpha+t}\xi^j_{\alpha+t}||_2^2 \\
&\le \frac{LK\gamma^2}{2M^2}\mathbb{E}\sum_{t=0}^{K-1}||\sum_{j=1}^{M}\nabla f(x^j_{\alpha+t}; D^j_t)||_2^2 + \frac{LK\gamma^2}{2M^2}\mathbb{E}\sum_{t=0}^{K-1}||\sum_{j=1}^{M}p^j_{\alpha+t}\xi^j_{\alpha+t}||_2^2 \\
&\le \frac{LK\gamma^2}{2M^2}\mathbb{E}\sum_{t=0}^{K-1}||\sum_{j=1}^{M}\nabla f(x^j_{\alpha+t}; D^j_t)||_2^2 + \frac{LK\gamma^2\check{p}^2}{2M^2}\sum_{t=0}^{K-1}\mathbb{E}||\sum_{j=1}^{M}\xi^j_{\alpha+t}||_2^2 \\
&\le \frac{LK\gamma^2}{2M^2}\mathbb{E}\sum_{t=0}^{K-1}||\sum_{j=1}^{M}\nabla f(x^j_{\alpha+t}; D^j_t)||_2^2 + \frac{LK^2\gamma^2\check{p}^2\sigma^2}{2M} \\
&\le \frac{LK^2\gamma^2\theta}{2M} + \frac{LK\gamma^2}{2}\sum_{t=0}^{K-1}\mathbb{E}||\nabla f(x^j_{\alpha+t})||_2^2 + \frac{LK^2\gamma^2\check{p}^2\sigma^2}{2M} \\
&= \frac{LK^2\gamma^2}{2M}(\theta + \check{p}^2\sigma^2) + \frac{LK\gamma^2}{2}\sum_{t=0}^{K-1}\mathbb{E}||\nabla f(x^j_{\alpha+t})||_2^2
\end{aligned}
$$

After getting the former and the latter derived results, we can substitute them into Equation 4.

$$\mathbb{E}f(\tilde{x}_{\alpha+1}) - f(\tilde{x}_\alpha) \leq -\frac{(K+1)\gamma}{2}(1 - \frac{L^2\gamma^2K(K-1)}{2(K+1)} - \frac{L\gamma K}{(K+1)})||\nabla f(\tilde{x}_\alpha)||_2^2$$
$$- \frac{\gamma}{2}(1 - \frac{L^2\gamma^2(K+1)(K-2)}{2} - L\gamma K)\sum_{t=1}^{K-1}\mathbb{E}||\nabla f(x_{\alpha+t}^j)||_2^2$$
$$+ (\frac{L^2\gamma^3(K-1)K(2K-1)}{12} + \frac{LK^2\gamma^2}{2M})(\theta + \check{p}^2\sigma^2)$$

Under the condition $1 \geq \frac{L^2\gamma^2(K+1)(K-2)}{2} + L\gamma K$,the second term on the right hand side can be discarded. This condition also implies that

$$\frac{(K+1)\gamma}{2}(1 - \frac{L^2\gamma^2K(K-1)}{2(K+1)} - \frac{L\gamma K}{(K+1)}) \geq \frac{\gamma(K - L^2\gamma^2)}{2}$$

Then, together with the condition $1 - \delta \geq L^2\gamma^2$ for some $0 < \delta < 1$,we have

$$\mathbb{E}f(\tilde{x}_{\alpha+1}) - f(\tilde{x}_\alpha) \leq -\frac{(K-1+\delta)\gamma}{2}||\nabla f(\tilde{x}_\alpha)||_2^2 + \frac{L\gamma^2K(\theta + \check{p}^2\sigma^2)}{2}(\frac{K}{M} + \frac{L(2K-1)(K-1)\gamma}{6})$$

By taking the summation we have

$$\mathbb{E}f(\tilde{x}_T) - f(\tilde{x}_1) \leq \sum_{j=1}^{T}\{-\frac{(K-1+\delta)\gamma}{2}||\nabla f(\tilde{x}_j)||_2^2 + \frac{L\gamma^2K(\theta + \check{p}^2\sigma^2)}{2}(\frac{K}{M} + \frac{L(2K-1)(K-1)\gamma}{6})\}$$

Under Assumption 2, we have

$$f^* - f(\tilde{x}_1) \leq f(\tilde{x}_T) - f(\tilde{x}_1)$$

Combining both, we can immediately get the following bound

$$\frac{1}{T}\mathbb{E}\sum_{j=1}^{T}||\nabla f(\tilde{x}_j)||_2^2 \leq [\frac{2(f(\tilde{x}_1) - f^*)}{T\gamma(K-1+\delta)} + \frac{L\gamma K(\theta + \check{p}^2\sigma^2)}{K-1+\delta}(\frac{K}{M} + \frac{L(2K-1)(K-1)\gamma}{6})]$$

Proof of Corollary

Proof. At first, we assume that $\frac{K}{M} > \frac{L(2K-1)(K-1)\gamma}{6}$, Then we can rewrite the bound as

$$\frac{1}{T}\mathbb{E}\sum_{j=1}^{T}||\nabla f(\tilde{x}_j)||_2^2 \leq [\frac{2(f(\tilde{x}_1) - f^*)}{T\gamma(K-1+\delta)} + \frac{2L\gamma K^2}{M(K-1+\delta)}(\theta + \check{p}^2\sigma^2)]$$

Set

$$f(\gamma) = \frac{2(f(\tilde{x}_1) - f^*)}{T\gamma(K-1+\delta)} + \frac{2L\gamma K^2}{M(K-1+\delta)}(\theta + \check{p}^2\sigma^2)$$

By taking $f' = 0$, one immediately get

$$\gamma = \sqrt{\frac{(f(\tilde{x}_1) - f^*)M}{K^2L(\theta + \check{p}^2\sigma^2)}} * \frac{1}{\sqrt{T}}$$

and

so

$$\frac{1}{T}\mathbb{E}\sum_{j=1}^{T}||\nabla f(\tilde{x}_j)||_2^2 \leq \sqrt{\frac{(f(\tilde{x}_1) - f^*)L(\theta + \check{p}^2\sigma^2)}{M}} * \frac{4K}{(K-1+\delta)\sqrt{T}}$$

By plugging in the value of $\gamma, \frac{K}{M} > \frac{L(2K-1)(K-1)\gamma}{6}$ implies that

$$T > \frac{((f(\tilde{x}_1) - f^*)LM^3(2K-1)^2(K-1)^2}{36K^4(\theta + \check{p}^2\sigma^2)}$$

At last, we need to check condition (3.1) is hold. It is sufficient to set

$$T > \frac{L(f(\tilde{x}_1) - f^*)M}{K^2(\theta + \check{p}^2\sigma^2)}$$

REFERENCES

[1] Fan Zhou and Guojing Cong. 2017. On the convergence properties of a K-step averaging stochastic gradient descent algorithm for nonconvex optimization. *arXiv preprint arXiv:1708.01012* (2017).